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Fundamental of optimization

Approximation algorithms

ONE LOVE. ONE FUTURE.

CONTENT

- Overview
- Knapsack problem
- TSP problem

- Algorithms with polynomial time complexity
- Find approximate solutions to optimization problems: the distances of the solutions found to optimal solutions can be probably guaranteed.
- Notation
 - Optimal objective value is f^*
 - Objective value of the solution found by the approximation algorithm is f
- Maximization problems
 - We can prove that: $f \geq \alpha f^*$ in which α is a positive constant less than 1
- Minimization problems
 - We can prove that: $f \leq \alpha f^*$ in which α is a positive constant greater than 1

- There are n items $A = \{1, 2, \dots, n\}$ in which item i has weight W_i and value C_i ($i = 1, 2, \dots, n$). Find a subset of A such that the sum of weights of items is less than or equal to B and the sum of values is maximal.
- Let S be a solution to the problem (S is a subset of A). Let $f(S) = \sum_{i \in S} C_i$ be the value of the solution S
- Let S^* be an optimal solution and $f^* = f(S^*)$ be the optimal objective value
- Consider 2 algorithms running on 2 sorted list of items:
 - Sort in non-increasing order of values
 - Sort in non-increasing order of the fractions of values over the weights

```
GREEDY-1 () {  
    ( $\{C_1, W_1\}, \{C_2, W_2\}, \dots, \{C_n, W_n\}$ ) is the sorted list in which:  $C_1 \geq C_2 \geq \dots \geq C_n$ ;  
     $S = \{\}$ ;  
    for  $i = 1$  to  $n$  do {  
        if  $W_i > B$  then break;  
         $S = S \cup \{i\}$ ;  $B = B - W_i$ ;  
    }  
    return  $S$ ;  
}
```

GREEDY-2 () {

$\{ \{C_1, W_1\}, \{C_2, W_2\}, \dots, \{C_n, W_n\} \}$ is the sorted list in which: $\frac{C_1}{W_1} \geq \frac{C_2}{W_2} \geq \dots \geq \frac{C_n}{W_n}$;

$S = \{\}$;

for $i = 1$ to n do {

 if $W_i > B$ then break;

$S = S \cup \{i\}$; $B = B - W_i$;

}

return S ;

}

- There are n items $A = \{1, 2, \dots, n\}$ in which item i has weight W_i and value C_i ($i = 1, 2, \dots, n$). Find a subset of A such that the sum of weights of items is less than or equal to B and the sum of values is maximal.
- **Approximation algorithms**
 - Let S_1 and S_2 be the solutions returned by GREEDY-1 and GREEDY-2
 - Notation: $f = \max\{f(S_1), f(S_2)\}$
 - We have $f \geq \frac{1}{2}f^*$

- Proof
- Consider the linear program below:

$$\begin{aligned} f(X) &= C_1X_1 + C_2X_2 + \dots + C_nX_n \rightarrow \max \\ \text{s.t. } W_1X_1 + W_2X_2 + \dots + W_nX_n &\leq B \\ 0 &\leq X_1, X_2, \dots, X_n \leq 1 \end{aligned}$$

KNAPSACK

- Simplex table

x_1	x_2	...	x_n	x_{n+1}	x_{n+2}	...	x_{2n+1}	Z	RHS
w_1	w_2	...	w_n	1	0	...	0	0	B
1	0		0	0	1		0	0	1
0	1	...	0	0	0	...	0	0	1
...	...	...	...	...	...	...	...	...	...
0	0		1	0	0		1	0	1
$-C_1$	$-C_2$	...	$-C_n$	0	0	0	0	1	0

KNAPSACK

	X_1	X_2	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS	E
R_1	W_1	W_2	...	W_n	1	0	...	0	0	B	B/W_1
R_2	1	0		0	0	1		0	0	1	1
R_3	0	1	...	0	0	0	...	0	0	1	∞
	...	...	...	...	...	...	...	...	...	...	
R_{n+1}	0	0		1	0	0		1	0	1	∞
R_{n+2}	$-C_1$	$-C_2$	...	$-C_n$	0	0	0	0	1	0	



Select column 1, suppose $B \geq W_1 \rightarrow$ select row R_2 : $R_1 = R_1 - W_1 R_2$; $R_{n+2} = R_{n+2} + C_1 R_2$

	X_1	X_2	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS	E
R_1	0	W_2	...	W_n	1	$-W_1$	...	0	0	$B - W_1$	
R_2	1	0		0	0	1		0	0	1	
R_3	0	1	...	0	0	0	...	0	0	1	
	...	...	...	...	...	...	...	...	...	...	
R_{n+1}	0	0		1	0	0		1	0	1	
R_{n+2}	0	$-C_2$	...	$-C_n$	0	C_1	0	0	1	C_1	

KNAPSACK

	X_1	X_2	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS	E
R_1	0	W_2	...	W_n	1	$-W_1$	...	0	0	$B - W_1$	$(B - W_1)/W_2$
R_2	1	0		0	0	1		0	0	1	∞
R_3	0	1	...	0	0	0	...	0	0	1	1
	...	...	...	...	...	...	...	...	...	...	...
R_{n+1}	0	0		1	0	0		1	0	1	∞
R_{n+2}	0	$-C_2$	...	$-C_n$	0	0	0	0	1	C_1	



Select column 2, Suppose $(B - W_1) \geq W_2 \rightarrow$ select row R_3 : $R_1 = R_1 - W_2 R_3$; $R_{n+2} = R_{n+2} + C_2 R_3$

	X_1	X_2	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS	E
R_1	0	0	...	W_n	1	$-W_1$	...	0	0	$B - W_1 - W_2$	
R_2	1	0		0	0	1		0	0	1	
R_3	0	1	...	0	0	0	...	0	0	1	
	...	...	...	...	...	...	...	...	...	...	
R_{n+1}	0	0		1	0	0		1	0	1	
R_{n+2}	0	0	...	$-C_n$	0	C_1	C_2	0	1	$C_1 + C_2$	

KNAPSACK

	X_1	X_2	X_3	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS	E
R_1	0	0	W_3	...	W_n	1	$-W_1$	...	0	0	$B - W_1 - W_2$	$(B - W_1 - W_2) / W_3$
R_2	1	0	0	...	0	0	1		0	0	1	∞
R_3	0	1	0	...	0	0	0	...	0	0	1	∞
R_4	0	0	1	...	0	0	0	...	...	...	1	1
R_{n+1}	0	0	0	...	1	0	0		1	...	...	∞
R_{n+2}	0	0	$-C_3$	...	$-C_n$	0	C_1	C_2	0	1	$C_1 + C_2$	



Select column 3, suppose $(B - W_1 - W_2) < W_3 \rightarrow$ select row R_1 : $R_1 = R_1 / W_3$; $R_{n+2} = R_{n+2} + C_3 R_1 / W_3$

	X_1	X_2	X_3	X_4	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS
R_1	0	0	1	W_4 / W_3	...	W_n / W_3	1	$-W_1$	...	0	0	$(B - W_1 - W_2) / W_3$
R_2	1	0	0	0	...	0	0	1		0	0	1
R_3	0	1	0	0	...	0	0	0	...	0	0	1
R_4	0	0	0	$-W_4 / W_3$	...	$-W_n / W_3$	0	0	...	...	...	...
R_{n+1}	..	..	..	...	...	1	0	0		1	0	1
R_{n+2}	0	0	0	$-C_4 + W_4 * C_3 / W_3$	...	$-C_n + W_n * C_3 / W_3$	0	C_1	C_2	0	1	$C_1 + C_2$

KNAPSACK

	X_1	X_2	X_3	X_4	...	X_n	X_{n+1}	X_{n+2}	...	X_{2n+1}	Z	RHS
R_1	0	0	1	W_4/W_3	...	W_n/W_3	1	$-W_1$	...	0	0	$(B-W_1-W_2)/W_3$
R_2	1	0	0	0	...	0	0	1		0	0	1
R_3	0	1	0	0	...	0	0	0	...	0	0	1
R_4	0	0	0	$-W_4/W_3$	...	$-W_n/W_3$	0	0	...	...	...	...
R_{n+1}	..	..	..	...	...	1	0	0		1	0	1
R_{n+2}	0	0	0	$-C_4+W_4*C_3/W_3$	...	$-C_n+W_n*C_3/W_3$	0	C_1	C_2	0	1	C_1+C_2

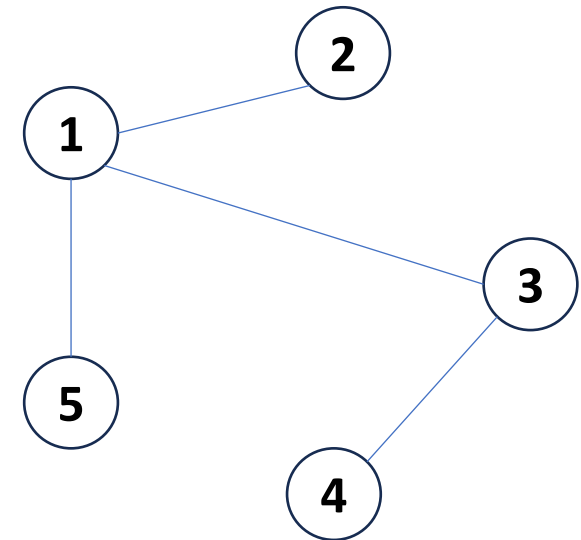
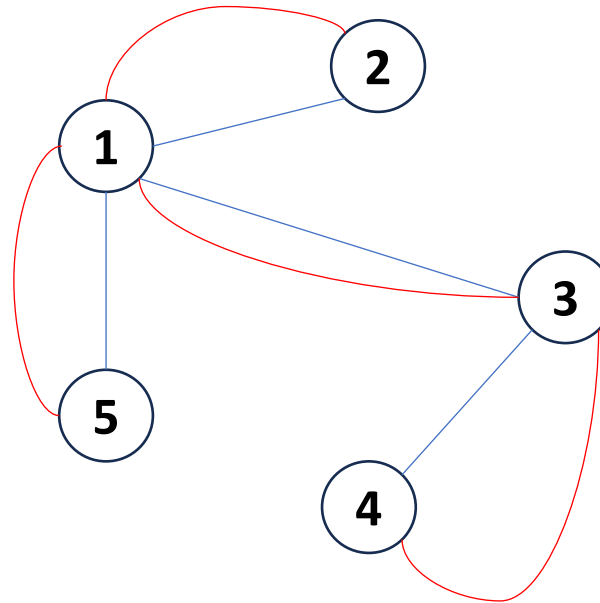
- Values of the last row are non-negative, the table corresponds to an optimal solution: $X_1 = 1, X_2 = 1, X_3 = (B-W_1-W_2)/W_3, X_4 = \dots = X_n = 0$.
- Generality: select column from left to right (from columns 1, 2, 3, ...). Suppose k is the first index (iteration) where $B-W_1-W_2-\dots-W_{k-1} < W_k$. An optimal solution to the linear problem is: $X_1 = X_2 = \dots = X_{k-1} = 1, X_k = (B-W_1-W_2-\dots-W_{k-1})/W_k, X_{k+1} = \dots = X_n = 0$ optimal objective value is $C_1 + C_2 + \dots + \alpha C_k$, với $\alpha = (B-W_1-W_2-\dots-W_{k-1})/W_k, (\alpha < 1)$

BÀI TOÁN KNAPSACK

- The given linear program is a relaxation of the original Knapsack problem: optimal objective value of the Knapsack problem is $f^* \leq C_1 + C_2 + \dots + \alpha C_k$. Objective value of the solution returned by the approximation is $f(S_1) = C_1 + C_2 + \dots + C_{k-1}$.
- We have $f^* \leq C_1 + C_2 + \dots + \alpha C_k < C_1 + C_2 + \dots + C_{k-1} + C_k = f(S_1) + C_k$. Therefore, $f(S_1) > f^*/2$ or $f(S_2) \geq C_k > f^*/2$.
- Hence $f = \max\{f(S_1), f(S_2)\} > f^*/2$.

TRAVELLING SALESMAN PROBLEM - TSP

- Let $G = (V, E)$ be a complete graph in which $V = \{1, 2, \dots, n\}$ is the set of nodes. Edge (u,v) has weight $c(u,v)$. Find the Hamilton cycle G such that the total weights is minimal.
- Approximation algorithm:
 - Let $T(G)$ be a minimum spanning tree of G .
 - Duplicate the edges of $T \rightarrow$ we obtain an Euler graph with the Euler cycle $E(T)$.
 - Let S^0 be the sequence (pairwise distinct) of nodes visited when travelling along $E(T)$. The addition of the first node of S^0 to the end of S^0 yields a Hamilton cycle S which is the solution of the approximation algorithm.
- Example
 - $E(T) = 1, 2, 1, 5, 1, 3, 4, 3, 1$
 - $S^0 = 1, 2, 5, 3, 4$
 - $S = 1, 2, 5, 3, 4, 1$



A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular shape composed of many small red dots. The dots are arranged in a way that creates a sense of depth and movement, resembling a spiral or a stylized 'H' shape. In the center of this graphic, the word 'HUST' is written in a bold, white, sans-serif font.

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THANK YOU !